



Vapor-Phase Deposited Polymer Dielectric Layers for Organic Electronics: Design, Characteristics, and Applications

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Abstract

The emergence of organic electronics has transformed the landscape of electronic devices, paving the way for future advancements in low-power, flexible, and wearable electronics compatible with various form factors. Polymeric dielectric layers are pivotal in the implementation of organic electronics due to their inherent deformable characteristics as well as outstanding insulating performance. Here, the review highlights an innovative technology termed initiated chemical vapor deposition (iCVD) for synthesizing polymer dielectric materials, particularly in the context of organic thin-film transistors (OTFTs). The all-dry polymer deposition process circumvents issues associated with conventional solvent-based methods, such as residual solvent, potential damage to the substrate, and the lack of large-area uniformity, allowing for ultra-thin, high-purity polymer dielectric layers with exceptional dielectric performance comparable to inorganic dielectrics. Furthermore, iCVD process enables the incorporation of various chemical functionalities into the dielectric layer, which enables the generation of versatile, high-performance organic electronic devices. Based on the beneficial aspects of the iCVD process, the review provides an overview of iCVD polymer dielectric layers, emphasizing their significance and potential toward innovative applications in the fields of organic electronic, including OTFTs, resistive random-access memory (RRAM), flash memory and logic circuits.

Keywords Polymer dielectric layer · Organic thin film transistor (OTFT) · Initiated chemical vapor deposition (iCVD) · Dielectric constant (k) · High- k dielectric · Stretchable dielectric layer · Resistive random-access memory (RRAM) · Organic flash memory · Multi-value logic (MVL)

Introduction

Since the 1970s, the Si-based semiconductor industry has been fundamental to the development of modern information society. Conventional complementary metal–oxide–semiconductor (CMOS) system is an integration comprising rigid

circuit boards and hard Si-based chips. They are not suitable to fulfill the requirement for flexibility in the forthcoming era of wearable electronics and soft robotics. The emergence of 'organic electronics' as a distinct category has revolutionized the field of electronic devices capable of extraordinary flexibility—enabling various types of mechanical deformation of the electronic devices including bending, folding, twisting, compression, and stretching while maintaining its electrical performance [1, 2].

Driven by polymer dielectrics and organic semiconductors, organic electronics have become a focal point in advancing next-generation electronic devices due to their advantageous properties such as low-temperature processability, lightweight, and inherent mechanical flexibility, rendering them compatible with plastic substrates. In the past 2 decades, there has been significant progress in the development of organic electronics thanks to the innovation in material synthesis, manufacturing technique, and

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device architecture [3, 4]. Various types of devices including non-volatile memory devices, multi-value logic devices, neuromorphic devices, and integrated circuits have been developed successfully based on organic thin film transistors (OTFTs).

One of the fundamental components to realize the aforementioned organic electronic devices is dielectric layer. For OTFTs, the dielectric layer acts as insulator between semiconductor layer and the gate electrode. It enables the control of charge carriers in the semiconductor channel when a voltage is applied to the gate, directly influencing the switching behavior of the device and overall performance characteristics such as charge carrier mobility (μ), threshold voltage (V_T), hysteresis, and leakage current. Given the critical role of the dielectric layer in organic electronics, various polymer materials have been successfully developed, including polystyrene (PS), polyimide (PI), poly(methyl methacrylate) (PMMA), polyurethane (PU), poly(vinyl alcohol), poly(4-vinylphenol) (PVP), poly(vinyl chloride) (PVC), and poly(vinylidene fluoride) (PVDF). All of these dielectric materials are solution-processable, and applied to various electronics by using spin-coating, bar-coating, dip-coating, spray-coating, or blade-coating. Unfortunately, it has been challenging to achieve thickness scaling down to nanometer scale while retaining pinhole-free films over large areas without compromising the robust insulating properties. Indeed, it was repeatedly observed that most of the solution-processed polymer dielectric layer lost the dielectric performance when the film thickness was lower than 100 nm.

Meanwhile, in last decade, a vapor-phase polymer deposition process, termed as initiated chemical vapor deposition (iCVD), was firstly proposed as an effective tool for synthesizing polymer dielectric layers, offering new avenues

for creating ultra-thin polymer dielectric layers with superb dielectric performance comparable to inorganic dielectrics [5–8]. Figure 1 illustrates the overall process flow of iCVD process. Similar to traditional free radical polymerization, the iCVD process involves free-radical reaction occurring in gas-phase, where vaporized initiator and monomers are introduced into a vacuum chamber and the polymerization reaction initiated by the radicals formed by thermal decomposition of initiator. Through the iCVD technique, polymer synthesis and film deposition arise concurrently on the substrate, producing highly uniform nano-scale polymeric films [9–11].

Beyond traditional solvent-based polymer synthesis, the iCVD process offers various advantages that are highly desirable for the applications in organic electronics. The vapor-phase process alleviates several issues associated with organic solvents. It enables the deposition of uniform polymer films with sub-10 nm thickness, which is quite challenging to achieve by a solution-based process, regardless of the substrate materials [12]. Additionally, it allows for homogeneous copolymerization, facilitated by the spontaneous mixing of monomer species in the gas phase. Moreover, through a sequential process, the damage-free interface formation between thin polymer films is easily achieved, ensuring excellent interfacial adhesion [13]. Chemical damage to the underlying substrate from solvent can be avoided by the solvent-free process. The substrate temperature in the iCVD was near room-temperature, which enables the direct deposition of various polymer thin films onto thermally vulnerable substrates, including organic semiconductors and elastomeric substrates [13, 14]. The vapor-phase process enables the fabrication of high-purity polymers without generating reaction byproducts during the iCVD process, which

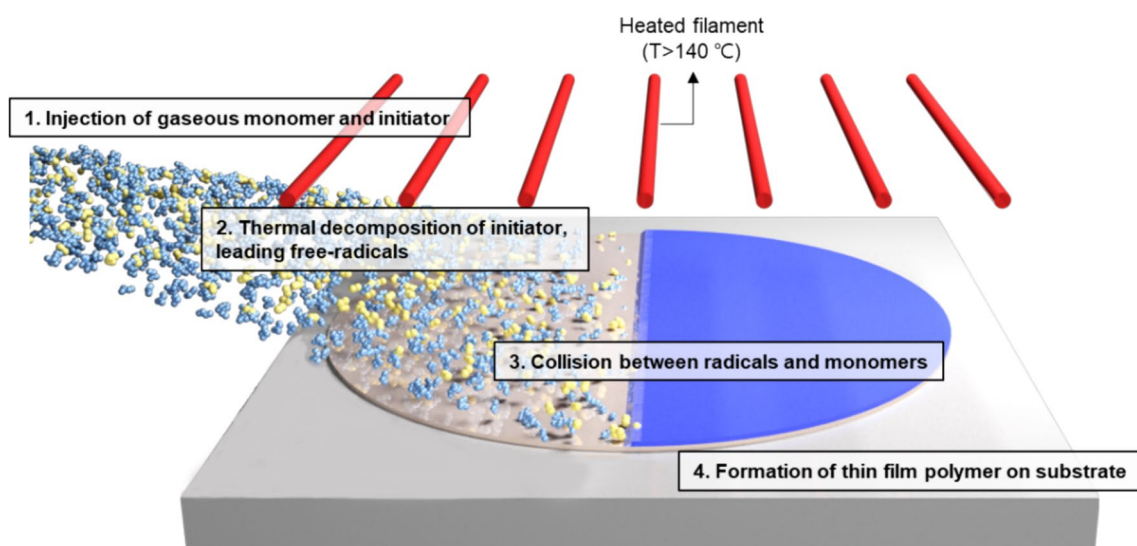


Fig. 1 Schematic illustration representing the iCVD process for polymer deposition

can otherwise lead to electrical leakage pathway [5]. Furthermore, the process guarantees essential features for very large-scale integration (VLSI), including high conformability, superb step coverage, and large-area uniformity [15–17].

Over a decade, various types of polymer dielectrics have been developed by iCVD process, as listed in Table 1. In this review paper, we cover most of the previously developed iCVD polymer dielectric layers and its array of applications. In more detail, in Section "Polymer gate dielectric layers for OTFTs", we discuss the role of iCVD polymer dielectric layer in thin film transistor (TFT), the essential component of electronic devices, highlighting the superior performance of vapor-deposited polymer dielectric layers. Then, we examine some innovative materials known for their electrical and mechanical excellence, including high- k and stretchable dielectrics. In Section "Organic electronic devices based on iCVD polymer dielectrics", we explore examples of electronic device applications beyond TFT utilizing iCVD dielectrics, such as resistive random-access memory (RRAM), non-volatile memory, and logic circuits. Specifically, we aim to delve into the performance of each device and the characteristics of the underlying materials.

Polymer Gate Dielectric Layers for OTFTs

OTFT is an essential component for future flexible/stretchable electronic products such as integrated circuits (ICs), radio-frequency identification tags (RFIDs), and sensors. To adopt OTFTs in these fields [25], it is crucial to develop high-performance materials for OTFTs. An OTFT includes three fundamental components: 1) an organic semiconductor (OSC) layer, 2) gate dielectric layer, and 3) conducting electrodes of gate (G), source (S), and drain (D). The dielectric layer plays a key role in OTFTs to maximize the charge flow in semiconductor as well as blocking unwanted charge flow between the semiconductor and gate electrode. For the operation of low-power OTFTs, achieving a high drain current (I_{DS}) is essential at a low applied gate voltage (V_G). The magnitude of source-drain current (I_{DS}) in the saturation region of an OTFT can be calculated using the equation as following.

$$I_{DS} = \left(\frac{W}{2L}\right)\mu_{\text{sat}}C_i(V_G - V_T)^2, \quad (1)$$

where W and L are the channel width and length, respectively, μ_{sat} is the saturation mobility, C_i is the capacitance per unit area, and V_G and V_T are the applied gate voltage and threshold voltage, respectively. The μ_{sat} and V_T values are closely related to the properties of channel materials and their interfaces [26]. According to the equation, to maximize the I_{DS} , a higher C_i value is imperative, which is determined by

$$C_i = \epsilon_0 \frac{k}{d}, \quad (2)$$

where ϵ_0 is the vacuum permittivity, k is the dielectric constant, and d is the thickness of the gate dielectric layer. Therefore, for high-performance OTFTs, materials exhibiting high dielectric constant while possessing excellent insulation characteristics are essential [27]. For this purpose, a thin-film deposition process enabling the conformal and uniform deposition of polymer dielectric layer over a large area are required.

iCVD Polymer Gate Dielectric Layers

Through the iCVD process, OTFT performance can be enhanced by controlling the aforementioned parameters. The solvent-free process enables polymer deposition without solvent-induced defects, thereby, leading to the formation of uniform dielectric layer with the thicknesses lower than 50 nm [28]. Since the substrate temperature is usually near room temperature—less than 40 °C in most cases, the process can be applied to various flexible organic substrates, such as poly(ethylene terephthalate) (PET), poly(ethylene naphthalate) (PEN), and PI [29–34]. This beneficial features render the iCVD process highly suitable for flexible electronics.

The poly(1,3,5-trimethyl-1,3,5-trivinyl cyclotrisiloxane) (pV3D3) is considered to a versatile polymer insulator for next-generation electronic devices due to the excellent insulating property, chemical and mechanical robustness, and easy processability to form a uniform, sub-10 nm-thick film synthesized by the iCVD process. The pV3D3 layer also showed good chemical stability owing to its high crosslinking density [35]. The film morphology and thickness remained unchanged even after 24 h of soaking into various organic solvents such as acetone and toluene, confirming its excellent chemical durability. Unprecedentedly outstanding insulating performance was observed from the pV3D3 dielectric layer, showing leakage current densities (J_i) lower than $10^{-8} \sim 10^{-9} \text{ A cm}^{-2}$ under electric field (E_i) more than 3MV/cm with sub-10 nm thickness of pV3D3 layer, which is unparalleled insulator performance among polymer insulators developed to date. This characteristic is even comparable to that of inorganic insulator such as thermally grown SiO_2 and high- k materials fabricated by atomic layer deposition (ALD) (Fig. 2a) [36, 37]. The charge transport mechanism of the pV3D3 was investigated through measuring the J_i – E_i characteristics of metal–insulator–metal (MIM) devices, as a function of device temperature (T_{dev}). The measured J_i – E_i characteristics match closely over a wide T_{dev} range from –140 to 10 °C (Fig. 2b), indicating that charge transport through the pV3D3 layer is governed mainly by tunnelling mode [38, 39]. It should be noted that

Table 1 List of iCVD dielectric

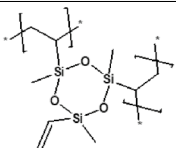
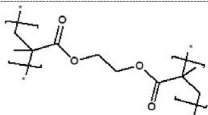
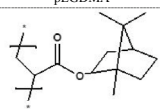
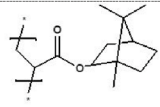
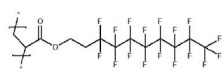
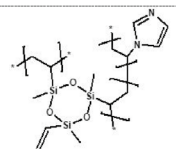
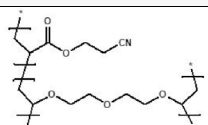
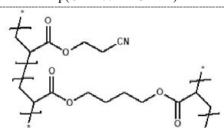
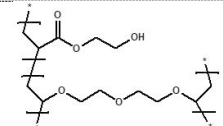
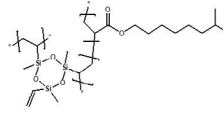
Structure (Name, abbreviation)	Property		Function and Application	Ref
	Breakdown voltage, E_{break} (MV cm ⁻¹)	k		
Flexible dielectric layers with outstanding insulating property				
 poly(1,3,5-trimethyl-1,3,5-trivinyl cyclotrisiloxane) pV3D3	> 5.5	2.2 – 2.3	- Gate insulator for flexible TFT - Low-k dielectric - Tunneling dielectric layer (TDL) - Resistive random-access memory (RRAM) - Organic encapsulation layer	[5, 19, 20]
 poly(ethylene glycol dimethacrylate) pEGDMA	> 6	3.02	- Gate insulator for flexible TFT - Resistive random-access memory (RRAM) - Blocking dielectric layer (BDL)	[6]
 poly(1,4-butanediol diacrylate) pBDDA	> 6	3.2	- Gate insulator for flexible TFT - Blocking dielectric layer (BDL)	[7, 8]
 poly(isobornyl acrylate) pIBA	> 4.2	2.61	- Gate insulator	[6]
 poly(3,3,4,4,5,5,6,6,7,7,8,8,9,9,10,10,10- -heptafluorodecylmethacrylate) pPFDMA	> 6	2.74	- Gate insulator - Low-k dielectric - Organic dielectric layer for triboelectric nano-generator (TENG)	[6]
 poly(1,3,5-trimethyl-1,3,5-trivinyl cyclotrisiloxane-co-1-vinylimidazole) p(V3D3-co-VI)	> 3.45	2.92 to 3.40 (k increases as the VI fraction increases)	- Gate insulator for flexible TFT - Metal cation chelating dielectric - Neuromorphic RRAM	[21]
High- k dielectric layers				
 poly(2-cyanoethyl acrylate-co- di(ethylene glycol) divinyl ether) p(CEA-co-DEGDVE)	0.6 to 4.3 (E_{break} increases as the DEGDVE fraction increases)	5.6 to 9.7 (k increases as the CEA fraction increases)	- High-k gate insulator for flexible TFT	[20, 22]
 poly(2-cyanoethyl acrylate-co-1,4- butanediol diacrylate) p(CEA-co-BDDA)	0.4 to 3.4 (E_{break} increases as the BDDA fraction increases)	5.6 to 9.7 (k increases as the CEA fraction increases)	- High-k gate insulator for flexible TFT	[23]

Table 1 (continued)

 poly(2-hydroxyethyl acrylate-co-di(ethylene glycol) divinyl ether) p(HEA-co-DEGDVE)	0.9 to 4.6 (E_{break} increases as the DEGDVE fraction increases)	6.2 to 8.9 (k increases as the HEA fraction increases)	- Charge trap layer (CTL) in non-volatile memory	[24]
Stretchable dielectric layers				
 poly(isononyl acrylate-co-1,3,5-trimethyl-1,3,5-trivinyl cyclotrisiloxane) p(INA-co-V3D3)	0.7 to 3.6 (E_{break} increases as the V3D3 fraction increases)	3.14 to 3.95 (k increases as the INA fraction increases)	- Stretchability (strain ~ 40%) - Stretchable TFT and logic circuit	[25]

such temperature-independent carrier conduction can be seen only when the defect density in the dielectric layer is sufficiently low so that thermally assisted conduction is suppressed significantly. This observation strongly supports that the iCVD process leads to the formation of polymeric films with exceptional purity and low defect density. It can be attributed partly to its solvent-free nature leaving no solvent residues in the final polymer films, thus making them less subject to formation of occasional pinholes, and to its unique film-forming mechanism, where the polymerization reaction is triggered by the dissociation of the initiator without damaging the input monomers [40–42].

The mechanical flexibility of the insulator is crucial for its application in flexible electronics, and pV3D3 layer is one of the best candidates for this purpose. As can be seen in the leakage characteristics of flexible MIM devices ($d_{\text{pV3D3}} = 19.6$ nm) prepared on 125 μm -thick PET substrates remained virtually unchanged on bending to a radius of curvature (R) as small as 1.5 mm for tensile stress and 2 mm for compressive stress. These values correspond to 4% and 3% for tensile and compressive strains, respectively (Fig. 2c). The tolerance of the pV3D3 layers to mechanical strain is significantly greater than most of inorganic insulating layers, which can hardly survive a tensile strain of up to 0.5% [43, 44]. Such a superior mechanical characteristic allows greater freedom in designing device architecture and choosing substrates and encapsulation layers for flexible electronics applications.

A series of other polymer dielectrics with excellent dielectric performance has also been synthesized including poly(ethylene glycol dimethacrylate) (pEGDMA), poly(isobornyl acrylate) (pIBA), and poly(1H, 1H, 2H, 2H-perfluorodecyl acrylate) (pPFDA) (see Table 1 and Fig. 2d). The k values of the pEGDMA, pIBA, and pPFDA are 3.02, 2.61, and 2.74, respectively. All polymer layers showed extremely low leakage current densities (J_i , of less than 10^{-8} A cm^{-2}) even at a thickness lower than 40 nm (Fig. 2e). OTFTs with pentacene and dinaphtho[2;3-b:2',3'-f]thieno [3,2-b]thiophene (DNNT) as p-type semiconductors

and with fullerene (C_{60}) as an n-type semiconductor were fabricated using the ultrathin iCVD polymer dielectric layers. Among them, OTFTs with pIBA exhibited the highest transistor performance. In addition to higher mobility levels (0.779 and 0.980 $\text{cm}^2\text{V}^{-1}\text{s}^{-1}$ for p-type DNNT and n-type C_{60} OTFTs, respectively), the lowest surface trap density levels (N_{ss}) were observed in the pIBA dielectric layer, although the surface energy and root-mean-square roughness (R_q) of pEGDMA and pIBA are similar with each other. This improved TFT performance presumably originates from surface polarities [45]. The ester group in pEGDMA is responsible for the increased surface polarities compared to pIBA. The excessive surface polarity in pEGDMA hinders the formation of a favorable interface with the following organic semiconductor, resulting in smaller grain organic semiconductor layer on the dielectric layer, resulting in the decrease in TFT performance [46]. This observation emphasizes the crucial role of iCVD polymer dielectric functionalities in the OTFT performance [23].

V_T is an importance parameter to lower power consumption and enhance the performance of OTFT [47]. V_T modulation had been achieved by using back or dual-gated electrodes, or self-assembled monolayer (SAM) treatment in the dielectric surface to generate electrostatic polarization at the dielectric-semiconductor interface [48]. However, these methods often complicated the TFT architecture or caused unwanted trapped charges and dipole moments at the interface, resulting in retardation of the charge carrier transport in the channel layer [49]. An alternative method was proposed for controlling V_T without altering the charge carrier mobility by adding the polar functionality in the bulk dielectric layer [20]. A series of copolymer dielectric layers (p(V3D3-co-VI) films) were synthesized from two monomers of 1,3,5-trimethyl-1,3,5-trivinyl cyclotrisiloxane (V3D3) and 1-vinyl-imidazole (VI). Electron-donating imidazole moiety in VI monomers can generate polarization effects to modulate V_T . pV3D3 synthesized by iCVD offers high levels of insulating properties and flexibility [5]. In addition, an ultrathin (3 nm) pV3D3 interface layer (IL)

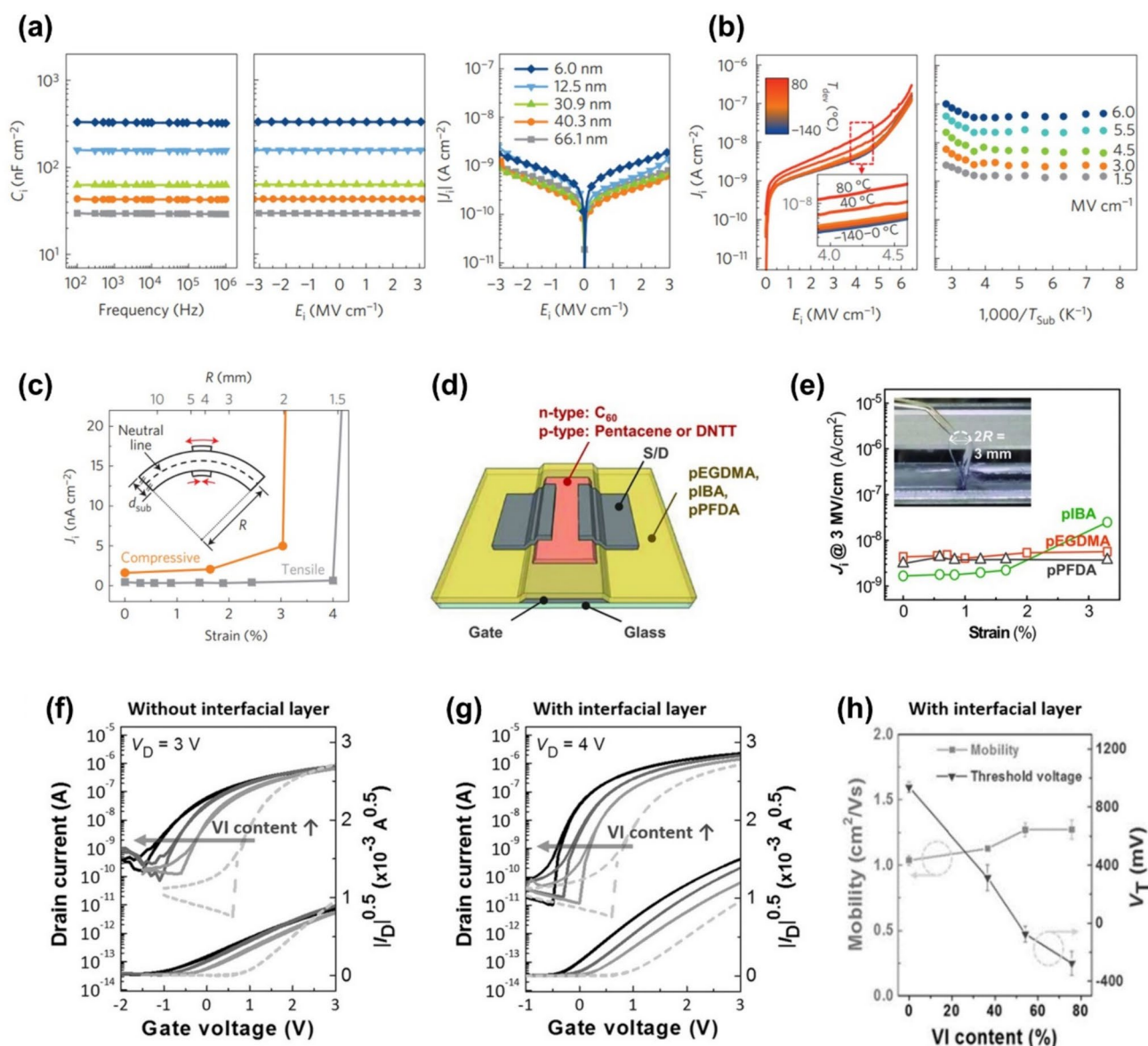


Fig. 2 Application of iCVD dielectric layer to organic thin-film transistors. **a** C_i versus frequency (left), C_i versus E_i (middle) and J_i versus E_i (right) measured for various d_{pV3D3} . **b** J_i - E_i characteristics of an Al/pV3D3/Al MIM device (d_{pV3D3} =6 nm) measured at various device temperatures (T_{dev}) (left) and the corresponding Arrhenius plots (right). **c** J_i measured at 3 MV cm⁻¹ for various values of bending radius (R) and corresponding strain (S) calculated using the relation of $S = d_{sub}/2R$, in which d_{sub} is the thickness of a substrate. Reproduced with permission from Ref. [5], Copyright 2015, Springer Nature. **d** Schematic of fabricated OTFTs with the iCVD dielectrics (pEGDMA, pIBA, pPFDA). **e** The J_i measured at 3 MV cm⁻¹ for

various tensile strain values for pIBA, pEGDMA, and pPFDA layers. The inset photograph shows the set-ups for bending test of MIM devices on PEN substrate. Reproduced with permission from Ref. [6], Copyright 2015, Wiley-VCH GmbH. **f** Transfer characteristics of C_{60} OTFTs using the series of copolymer dielectrics without the interfacial layer. **g** Transfer characteristics of C_{60} OTFTs using the series of copolymer dielectrics with the interfacial layer. The arrow denotes the increasing VI fraction. **h** Mobility (left axis) and V_T (right axis) as a function of VI content in the copolymer dielectrics with the interfacial layer. Reproduced with permission from Ref. [20], Copyright 2016, Wiley-VCH GmbH

was introduced between semiconductors and copolymer dielectrics to minimize the charge scattering in the OTFT channels regardless of the surface polarity of the copolymer. Bottom-gate C_{60} OTFTs were fabricated by implementing the copolymer dielectrics with and without IL to monitor the V_T shift of the device (Fig. 2f, g), which showed that

the V_T of C_{60} TFT with 3 nm-thick IL could be tuned systematically toward negative direction (ranging from 0.94 to -0.28 V) without changing charge carrier mobility (higher than 1 cm²V⁻¹ s⁻¹), as shown in (Fig. 2h). It follows from the observations above that the performance of OTFTs can be optimized systemically by incorporating various chemical

functionalities into the gate dielectric layer while keeping the dielectric thickness less than 30 nm, which is extremely challenging to achieve with conventional solution-based dielectric coating processes.

iCVD High- k Polymer Dielectrics

As mentioned in Eq. 2, dielectric with high dielectric constant is advantageous as it increases C_i values, allowing for higher I_{DS} with low voltages [50]. OTFTs with low driving voltage can be used in various applications such as wearable electronics, sensor arrays and flexible displays [51, 52]. A wide variety of high- k flexible polymer dielectrics have been studied in OTFT technology [53, 54]. The high- k dielectric can be formed through the copolymerization of the two monomers, one containing a polar functional group and the other with excellent insulating property. In Fig. 3a, the structural formula representing the polymer synthesis process via iCVD process are depicted.

Among the promising high- k monomer candidates, 2-cyanoethyl acrylate (CEA) can possess high dielectric constant of ~ 12.8 when polymerized, due to its polar cyano group (-CN) (Fig. 3b) [55–58]. However, the insulator performance

of cyanide-containing polymer film was extremely poor and showed effective insulating performance only with the thickness above 500 nm, which substantially lowered the C_i value consequently [59]. To address this issue, Choi et al. introduced a crosslinker di(ethylene glycol) divinyl ether (DEGDVE) to improve poor dielectric property of pCEA homopolymer. Notably, this is the first report on vapor-phase synthesis of a polymer with cyano group possessing versatile thickness adjustability. By tuning the input ratio of monomers, the composition of copolymer was optimized to generate pC1D1, which showed a high- k value of 6.2 and the breakdown voltage (E_{break}) greater than 3 MV/cm. The pC1D1 copolymer film was also equipped with outstanding mechanical flexibility on PEN substrates and chemical stability in various polar solvents [21]. Even though the addition of crosslinker is inevitable to enhance the insulating performance, the reduction in the dielectric constant of the copolymer dielectric layer must be minimized. For this purpose, other type of crosslinker, 1,4-butanediol divinyl ether (BDDVE) with shorter crosslinking chain length than DEGDVE was employed to mitigate the decrease in dielectric constant caused by the crosslinker with larger free volume (Fig. 3c). Precise control of chemical composition of

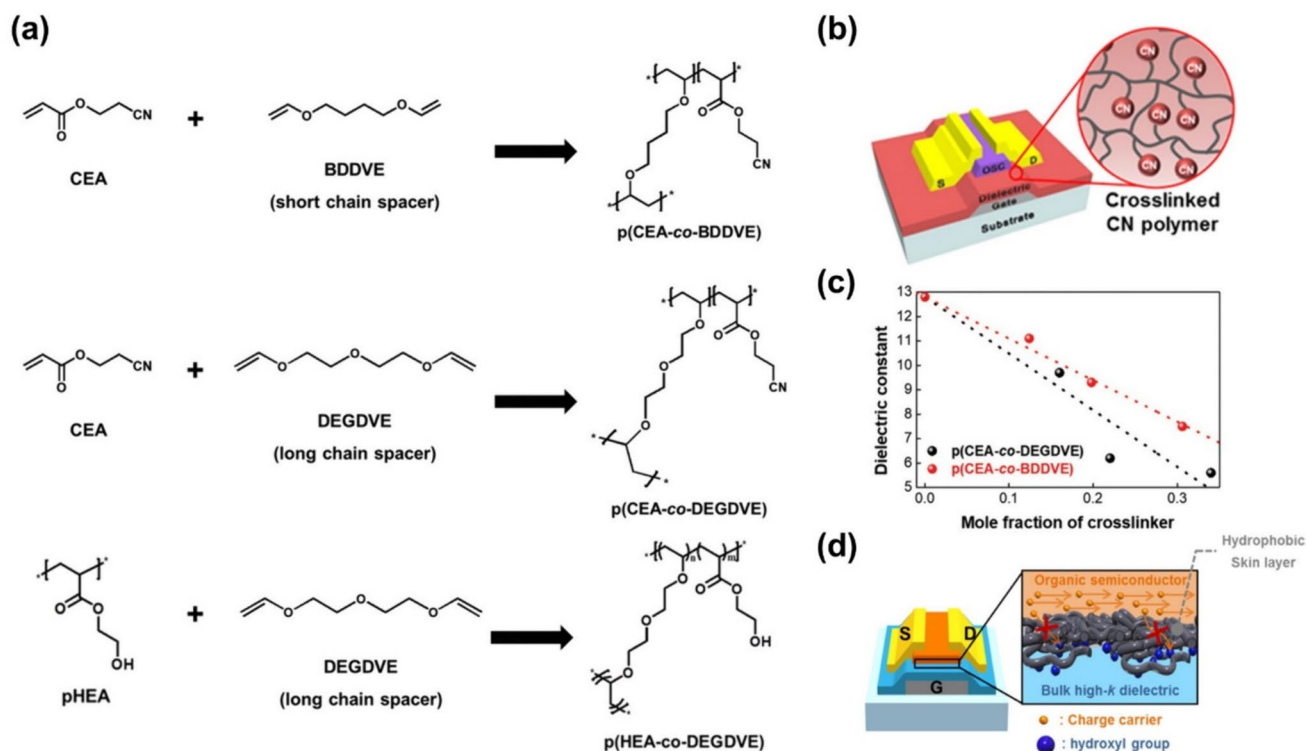


Fig. 3 Design of high- k , ultrathin polymer dielectrics synthesized by iCVD method. **a** Chemical structures of high- k copolymers p(CEA-co-DEGDVE), p(CEA-co-BDDVE), and p(HEA-co-DEGDVE). **b** Image of crosslinked high- k polymer dielectric containing highly polar cyano groups. Reproduced with permission from Ref. [21], Copyright 2017, American Chemical Society. **c** Estimation of dielec-

tric constant of CEA copolymer dielectrics varying mole fraction of crosslinker DEGDVE and BDDVE. Reproduced with permission from Ref. [22], Copyright 2020, Wiley-VCH GmbH. **d** Image of molecular thin hydrophobic skin layer of bulk high- k dielectric enabled by all-dry, vapor-phase process. Reproduced with permission from Ref. [23], Copyright 2019, American Chemical Society

CEA and BDDVE demonstrates record-high k value of 7.5 at a thickness below 30 nm. Flexible OTFTs using p(CEA-co-BDDVE) dielectrics maintain excellent insulating performance under a high tensile strain of 2.44%, positioning the high- k , ultrathin polymer dielectrics as promising candidates for future flexible electronics [22]. In addition to cyano groups, hydroxyl groups (-OH) are also considered as a promising polar functionality capable of exhibiting high- k values [60]. However, the highly polar hydroxyl groups present on the surface of the dielectric can cause severe hysteresis through trapping of charge carriers [61–64]. To mitigate this issue, a set of p(HEA-co-DEGDVE) copolymer dielectrics with polar hydroxyl groups was synthesized. The all-dry synthesis environment facilitated generation of a molecular thin hydrophobic surface skin layer, contributing to excellent organic semiconductor/dielectric interface properties in OTFTs (Fig. 3d). The dielectric with the optimum composition exhibited a high dielectric constant of 6.2 and extremely low leakage current with the dielectric thickness below 20 nm [23]. Such iCVD-synthesized ultrathin high- k copolymer dielectrics showed outstanding insulating performance and operational stability, showing a potential choice for future flexible electronic devices.

iCVD Stretchable Dielectrics

Recent advancements in wearable and implantable electronics ultimately require the stretchability of the electronic devices for full compatibility with various architectural form factors to achieve conformal contact with soft human tissue without compromising the performance degradation under mechanical deformation [65–71]. From the last decade, various approaches have been explored to impart stretchable characteristics to electronic devices, such as pre-stretching techniques, the use of rigid island-serpentine interconnect designs [72–84], or the adoption of intrinsically stretchable materials [85]. Compared to the methods based on geometrically designed rigid components, the use of intrinsically stretchable materials offers fundamental advantages on mechanical compliance, directionless deformability, and high-density integration capabilities [65, 86–88]. However, the fabrication of such devices faces various daunting challenges in all fabrication processes, including new materials synthesis, integration processes, lithographic patterning, and large-area integration.

Specifically, a significant technological challenge in intrinsically stretchable electronic circuits is the lack of stretchable dielectric materials with superior insulator performance. Moreover, materials possessing inherent stretchability often exhibit susceptibility to organic solvents, and thus are not compatible with standard microfabrication processes. Such obstacles critically limited the seamless fabrication of integrated systems. Despite the

recent advancement in intrinsically stretchable electronics [66], there still remains a crucial requirement for scalable production of a dielectric material that is ultrathin, highly stretchable, and retaining superior dielectric performance. The ideal stretchable dielectric layer should meet various requirements, including minimizing the dielectric thickness for low-voltage operation [27], forming a smooth interface for efficient charge transport, maintaining high insulation performance during stretching, ensuring high uniformity over large areas, and compatibility with microfabrication processes. However, previously reported intrinsically stretchable transistors often utilized solution-processed elastomeric dielectrics with drawbacks such as micrometer-scale thickness and low dielectric constants [70], resulting in high operation voltage and low gate controllability.

To address this issue, Koo et al. reported a stretchable TFT array with carbon nanotube channel and microcracked gold electrodes, the elastomeric insulating layer of p(INA-co-V3D3) was utilized as a gate dielectric layer (Fig. 4a). Isononyl acrylate (INA), characterized by its acrylic nature and extended alkyl chain, serves as a stretchable component within the polymer composition [24]. In its polymer state, it functions as a soft segment. On the other hand, V3D3 acts as a hard segment capable of crosslinking, thereby imparting the structural integrity as well as the restoring force to the polymer dielectric. The stretchable polymer network composed of p(INA-co-V3D3), using the INA monomer for softness and V3D3 as a crosslinker was synthesized via the iCVD process (Fig. 4b).

The 200 nm-thick organic network exhibited robust dielectric properties, in conjunction with notable stability against both mechanical and chemical stresses. An increase in soft INA segments resulted in an earlier breakdown (Fig. 4c). Although E_{break} of pI2V1 was 0.7 MV cm^{-1} , enhancements were observed in the E_{break} values for pI1V2 reaching 3.6 MV cm^{-1} . This improvement was attributed to the increased crosslinking density from a higher fraction of V3D3. The J_i - E_i curves were obtained for various dielectric thicknesses using the MIM structure with pI1V1 (Fig. 4d). Even with an extremely thin dielectric layer (185 nm), J remained below $1.0 \times 10^{-8} \text{ A cm}^{-2}$ within the range of $\pm 1 \text{ MV cm}^{-1}$. The capacitance of the MIM structure with a 160-nm-thick pI1V1 film was measured under various strains (Fig. 4e). The initial C_i before stretch was approximately 0.39 nF at a measured frequency of 1,000 Hz. After stretching to 40% strain, the C_i increased to 0.45 nF, measured at the same frequency. The 200 nm-thick dielectric layer could evenly cover an 8-inch wafer, creating a conformal and smooth interface with the carbon nanotube network. The transistors with the stretchable dielectric exhibited electrical performance comparable to conventional inorganic dielectrics such as aluminum oxide. Importantly, these transistors based on iCVD maintained uniform electrical performance

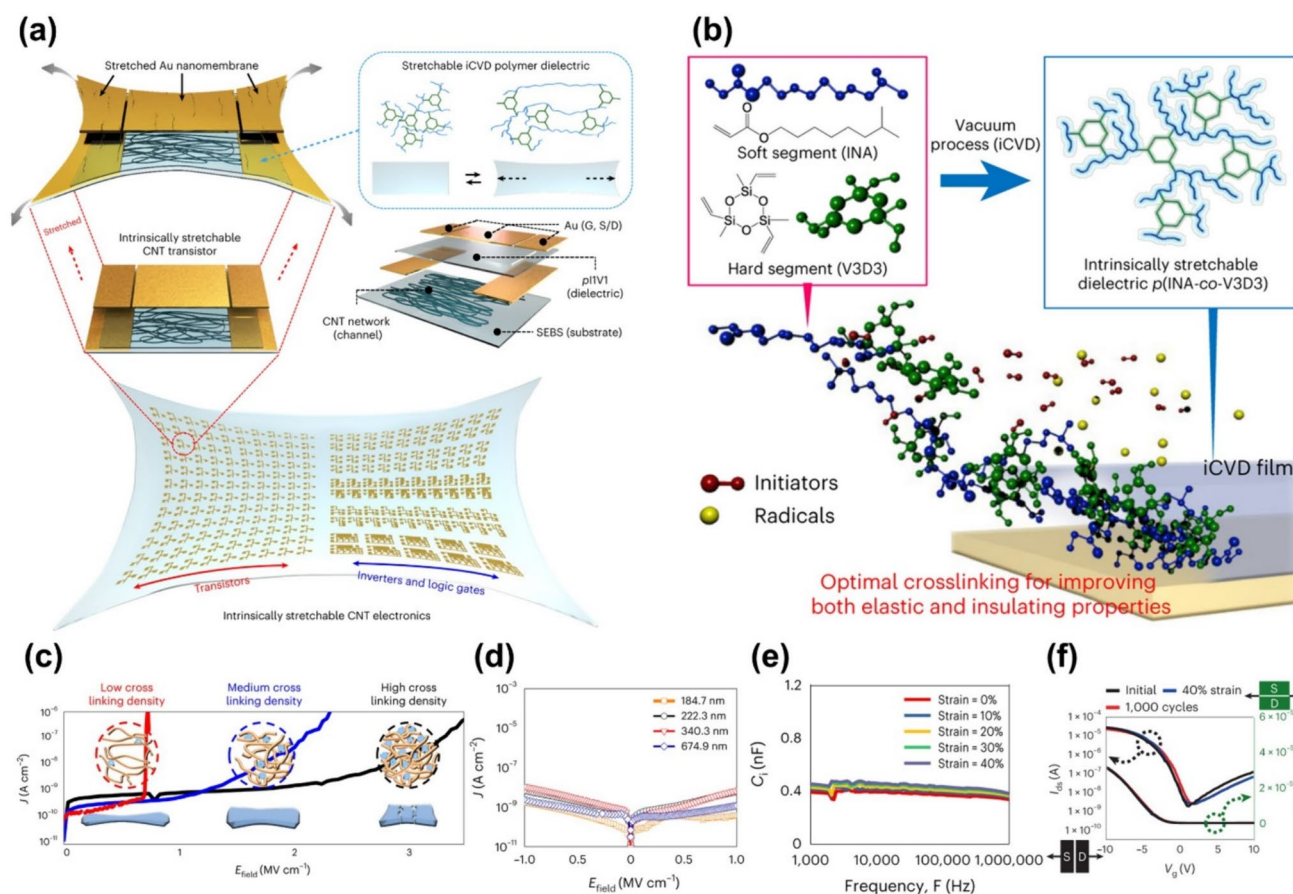


Fig. 4 Design of devices and dielectric materials employing iCVD covalently crosslinked networks. **a** Schematic of the stretchable electronic devices, incorporating inherently stretchable transistors with a channel formed from spin-coated s-CNT networks, ultrathin polymer dielectric layer, and microcracked gold thin film used for the gate, source, and drain electrodes. **b** Schematic of utilizing the iCVD process to deposit a stretchable polymer dielectric, involving INA and V3D3 monomers. **c** J - E_{field} properties of stretchable dielectrics with different monomer compositions. **d** J - E_{field} curves of MIM

devices with pIIV1 of different thicknesses. **e** C_i - F curves of MIM devices with pIIV1 dielectric under different strains. **f** Testing the intrinsically stretchable CNT transistors under stretching conditions. Transfer curves measured prior to stretching, during stretching with a 40% strain, and following 1000 cycles of repetitive stretching with a 40% strain in both parallel and perpendicular directions to the charge transport axis. Reproduced with permission from Ref. [24], Copyright 2023, Springer Nature

throughout the array and remained stable even after 1000 times of repeated stretching cycles at a 40% strain (Fig. 4f). Furthermore, simple level of circuits including inverters and logic gates were also produced with the copolymer dielectric layers, and operated effectively even at 40% applied strain.

Organic Electronic Devices Based on iCVD Polymer Dielectrics

Resistive Random-Access Memory (RRAM)

RRAM is a non-volatile memory with a simple MIM structure. When an electric field is applied to the electrode, the resistance of the insulator changes, which can be utilized to store and read data [89, 90]. RRAM provides various

advantages, including simple fabrication procedure, excellent scalability, fast operation speed, long data retention, and compatibility with existing CMOS technology. Recent advances in RRAM devices enable artificial intelligence (AI) tasks such as big data analysis [91], speech/face recognition [92], and healthcare data management [93]. To enhance the task efficiency, a growing interest can also be found in hardware-based brain-inspired neuromorphic systems [94]. These systems mimic the parallel networks of the brain, offering a promising potential for minimal energy consumption. The development of scalable, low-power electronic synapses is crucial for constructing large-scale, energy-efficient neuromorphic systems. In particular, organic RRAM can be utilized in flexible and wearable memory devices owing to its low processing temperature compatibility with plastic substrates, leveraging versatile

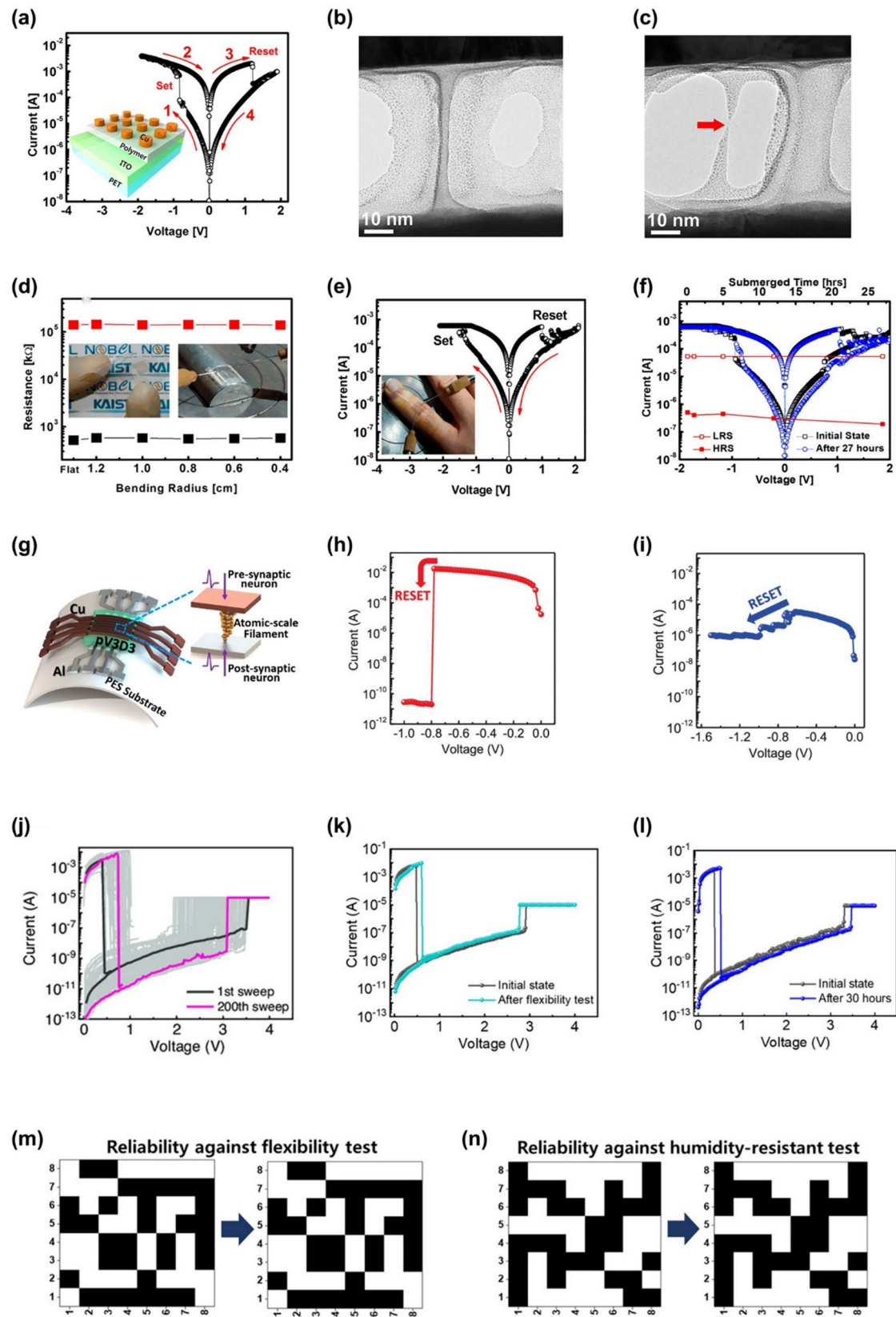


Fig. 5 Various structures and characteristics of RRAM utilizing iCVD dielectric films. **a** I - V characteristics of pEGDMA RRAM with bipolar switching mode. The inset image indicates a schematic of pEGDMA RRAM. **b** TEM image of the carbon filament formed for LRS of pEGDMA RRAM. **c** TEM image of the carbon filament abrupt for HRS of pEGDMA RRAM. **d** Changes in the HRS and LRS concerning the bending radius, indicating the flexibility of pEGDMA RRAM. **e** I - V characteristics during memory operation on a finger with flexible pEGDMA RRAM. **f** I - V characteristics and the HRS and LRS current of pEGDMA RRAM as in relation to the duration of submersion, representing the outcomes assessed in water. Reproduced with permission from Ref. [101], Copyright 2015, American Chemical Society. **g** Structure of flexible pV3D3 RRAM as electronic synapse. **h** Reset performance of the flexible pV3D3 RRAM following the creation of a Cu filament using a I_{cc} of 10^{-5} A. **i** Reset performance of the flexible pV3D3 RRAM with a thin filament produced through the gradual I_{cc} increase approach. Reproduced with permission from Ref. [106], Copyright 2019, American Chemical Society. **j** Initial and 200th I - V characteristics of the flexible pV3D3 RRAM observed through repetitive DC voltage sweeps. **k** I - V characteristics of the flexible pV3D3 RRAM before and after bending. **l** I - V characteristics of the flexible pV3D3 RRAM before and after the submerged duration of 30 h in water. Reliability indicated by intra-HD of pV3D3 RRAM in response to both **m** flexibility test and **n** humidity-resistant test. Reproduced with permission from Ref. [112], Copyright 2022, Wiley-VCH GmbH

organic materials with good mechanical strength [95–100]. Since most organic RRAM devices have been fabricated by solution-based processes, various inevitable problems such as substrate damage, low yield and poor film conformability had not yet been addressed properly [40].

Lee et al. aimed to resolve the shortcomings of organic RRAM by introducing iCVD pEGDMA dielectric [101]. The RRAM based on pEGDMA dielectric layer (pEGDMA-RRAM) operated at a low operating voltage around ± 1 V, and possessed a substantial memory window of over 10^2 (Fig. 5a). The pEGDMA-RRAM underwent a switching operation through the formation of a carbon filament (CF). When a bias was applied to the electrode, Joule heating was induced to the pEGDMA insulating layer to cause partial breakdown at the weakest point in the insulator [102]. As the applied bias intensity increased, the breakdown accelerated and altered the conductivity of the initial polymer by pyrolysis. Upon reaching the set voltage, an ultra-narrow but highly conductive CF bridged the top electrode (TE) and bottom electrode (BE), constituting the low-resistance state (LRS). The additional thermal energy induced the rupture of the CF, causing the RRAM to return to the high-resistance state (HRS). Figure 5b, c represent the formation of the CF in the LRS and the disruption of the CF in the HRS of pEGDMA-RRAM with the first clear illustration of the CF formation, and thus the operating mechanism of CF-based organic RRAM [103]. Furthermore, pEGDMA-RRAM exhibited stable operation even after more than 10,000 repetitive programming–erasing cycles with a bending radius of 4 mm (Fig. 5d), verifying its powerful potential as a flexible

memory for the internet of things (IoT) applications (Fig. 5e) [104, 105]. Also, thanks to the excellent chemical durability of the pEGDMA layer, pEGDMA-RRAM demonstrated a substantial memory window ($> 10^2$), maintaining retention for over 10^6 s even after being immersed in water for longer than 27 h (Fig. 5f).

Jang et al. employed RRAM based on pV3D3 (pV3D3-RRAM), utilizing a Cu TE and an Al BE, for use in face recognition neuromorphic systems (Fig. 5g) [106]. The pV3D3-RRAM relied mainly on the formation and rupture of a Cu conducting filament for its memory operation, which was different from aforementioned CF-based pEGDMA-RRAM. They focused on the transition of the reset behavior of pV3D3-RRAM from abrupt mode to gradual mode by varying the compliance current (I_{cc}), because analog conductance update is crucial for neuromorphic operation. Unlike the reset behavior observed when I_{cc} is fixed at 10^{-5} A (Fig. 5h), the pV3D3-RRAM exhibited a shift in reset behavior within the negative bias region. It transitioned to gradual switching mode with a reduced reset current by gradually increasing I_{cc} from 10^{-8} to 10^{-6} A (Fig. 5i).

Besides artificial intelligence (AI)-related tasks such as face recognition, security systems have also gained a huge attention due to cybercrime issues. Physically unclonable functions (PUFs) represent one of the most promising hardware security primitives due to their inherent clone-proof nature, cost-efficiency, and strong immunity to physical attacks [107–110]. Among the PUFs, RRAMs have been particularly appealing for their ability to provide stochastic in-memory computing capability and their straightforward structure suitable for cross-bar array matrices [111]. Building on the suggested pV3D3-RRAM mentioned above, Oh et al. demonstrated a pV3D3-based PUF (pV3D3-PUF) for highly secure neuromorphic systems [112]. Thanks to the outstanding mechanical deformability, pV3D3-PUF exhibited stable operation even after more than 200 cycles of repetitive voltage sweeps (Fig. 5j). Furthermore, pV3D3-PUF maintained stable memory operation even after 1000 repeated bending cycles with a 2.6% strain (Fig. 5k). Additionally, owing to the excellent chemical durability of pV3D3, pV3D3-PUF demonstrated stable operational characteristics even after being immersed in water for over 30 h (Fig. 5l). The exceptional flexibility and resistance to humidity of pV3D3-PUF warranted no instances of failure bits during PUF operation, ensuring the superior reliability of the pV3D3-PUF for applying security systems in IoT applications (Figs. 4m and Fig. 5n).

Organic Flash Memory

Flash memory is an essential non-volatile memory device required practically in all electronic systems [113]. To ensure stable data storage in wearable devices that undergo

repeated mechanical deformation, it is necessary to develop organic flash memory to overcome the limitation in the rigidity of conventional CMOS-based flash memory [114]. The structure of organic flash memory resembles that of the OTFTs, except for the gate dielectric layer divided into three layers: The organic flash memory includes a floating gate (FG) for charge storage, surrounded by a blocking dielectric layer (BDL) and a tunneling dielectric layer (TDL). By applying a voltage pulse (V_{prg}) to the control gate (CG), charges can tunnel through the TDL between the semiconductor and the FG [115–117]. Depending on the direction of charge movement, the memory operation is called either ‘programming’ (or ‘writing’) or ‘erasing’. When charges are stored in the FG, a built-in potential is formed within the flash memory, making V_T shift of the device. Such a V_T shift differentiates the ‘on current’ and ‘off current’ under the same reading voltage, indicating whether information is stored or not [118].

As the thickness of TDL determines the magnitude of the V_{prg} , nanoscale ultra-thin TDL is highly desirable for low-power memory operation. At the same time, the BDL and TDL must possess indestructible insulating properties to retain the stored charges in the FG for quite a long periods [117]. However, it is not trivial to realize polymer dielectric layers retaining both ultra-low thickness and robust insulating performance. The polymer TDL and BDL fabricated by the iCVD process are only few dielectric materials that can satisfy the requirements for organic flash memory, enabling the realization of high-performance flexible organic electronic systems.

Lee et al. utilized the iCVD process for both TDL and BDL, successfully fabricating high-performance, low-voltage flash-memory on various flexible substrates [119]. The organic flash memory device was composed of pV3D3 TDL and pEGDMA BDL with Al FG (Fig. 6a, b). Thanks to the nanoscale thickness and superior insulating characteristics of the pV3D3 polymer dielectric, the TDL effectively facilitated Fowler–Nordheim (F–N) tunneling even at low voltage [120]. The pEGDMA was chosen for BDL due to high breakdown voltage exceeding 6 MV/cm, effectively preventing charge leakage between FG and gate electrode. The organic flash memory showed a wide memory window of 25 V at operating voltages under 10 V, and high retention characteristics up to $\sim 10^8$ s. The organic flash memory retained its performance even after significant bending and folding cycles on various flexible substrates, including paper and 6 micron-thick Mylar™ (Fig. 6c).

To accomplish programming and erasing the organic flash memory effectively at low voltage ranges, it is crucial to increase the coupling ratio of TDL and BDL [121]. A higher coupling ratio increases the ratio of applied electric field across the TDL when a voltage is applied to the CG, effectively facilitating F–N tunneling even under low V_{prg} .

By introducing a high- k BDL, the coupling ratio can be increased substantially [122–124]. Woo et al. developed a flash memory using an MoS₂ channel with low- k ($k \sim 2.2$) pV3D3 TDL and high- k ($k \sim 5.4$) Al₂O₃ BDL (Fig. 6d, e) [125]. By using high- k Al₂O₃ and low- k pV3D3, they increased the effective electric field on the TDL, achieving memory window of 5.2 V with the programming voltage as low as 13 V along with excellent retention time over 10^5 s and endurance more than 1000 cycles. The iCVD process ensured a smooth, pinhole-free interface between pV3D3 and MoS₂, overcoming the challenges of solution process in conformal contact, leading to a tunable memory window with a high on/off ratio of 10^6 (Fig. 6f).

Conventional flash memory with metal FGs provides long retention times but involves a complex fabrication process with repeating metal and non-metal layers, and inevitable involvement of unwanted parasitic capacity [126]. When a polymer dielectric is used as a charge-trapping layer (CTL), it not only addresses the aforementioned issues but also enhances the stability of the flash memory device against mechanical deformation [127].

Lee et al. fabricated a flash memory device by replacing FGs with a polymer CTL [7]. They utilized pV3D3 for TDL and p(1,4-butanediol diacrylate) (pBDDA) for BDL and poly(2-hydroxyl acrylate-co-1,4-butanediol diacrylate) [p(HEA-co-BDDA)] copolymer for the CTL (Fig. 6g). They successively deposited 30 nm-thick tri-layer polymer dielectrics by the iCVD process in a single batch. The hydroxyl-containing HEA moiety in CTL could trap and stabilize electrons, rendering p(HEA-co-BDDA) dielectric an effective CTL [61]. With higher HEA ratio, the organic flash memory showed superior retention property ($\sim 10^8$ s) (Fig. 6h). On a flexible PEN substrate, the memory devices still maintained robust memory characteristics under 2.72% tensile stress (Fig. 6i). By fabricating flash memory device with a 30 nm-thick tri-layer polymer dielectrics, the organic flash memory could show efficient, low-power memory device with exceptional retention time.

Applications on the Logic Devices

Organic semiconductors can function as either p-type or n-type semiconductors without the doping process, which is a significant advantage for constructing CMOS-based complementary logic circuit as it utilizes both p-type and n-type semiconductors [3, 128–133]. Furthermore, the molecular engineering of the organic semiconductors allows for the optimized electrical characteristics, such as V_T [5], charge mobility [134], and energy band structure [135], which is highly desirable for achieving complementary inverters with balanced p-type and n-type performance [4, 136–139]. Unfortunately, the low stability of organic semiconductors in solvents and high-temperature environment render

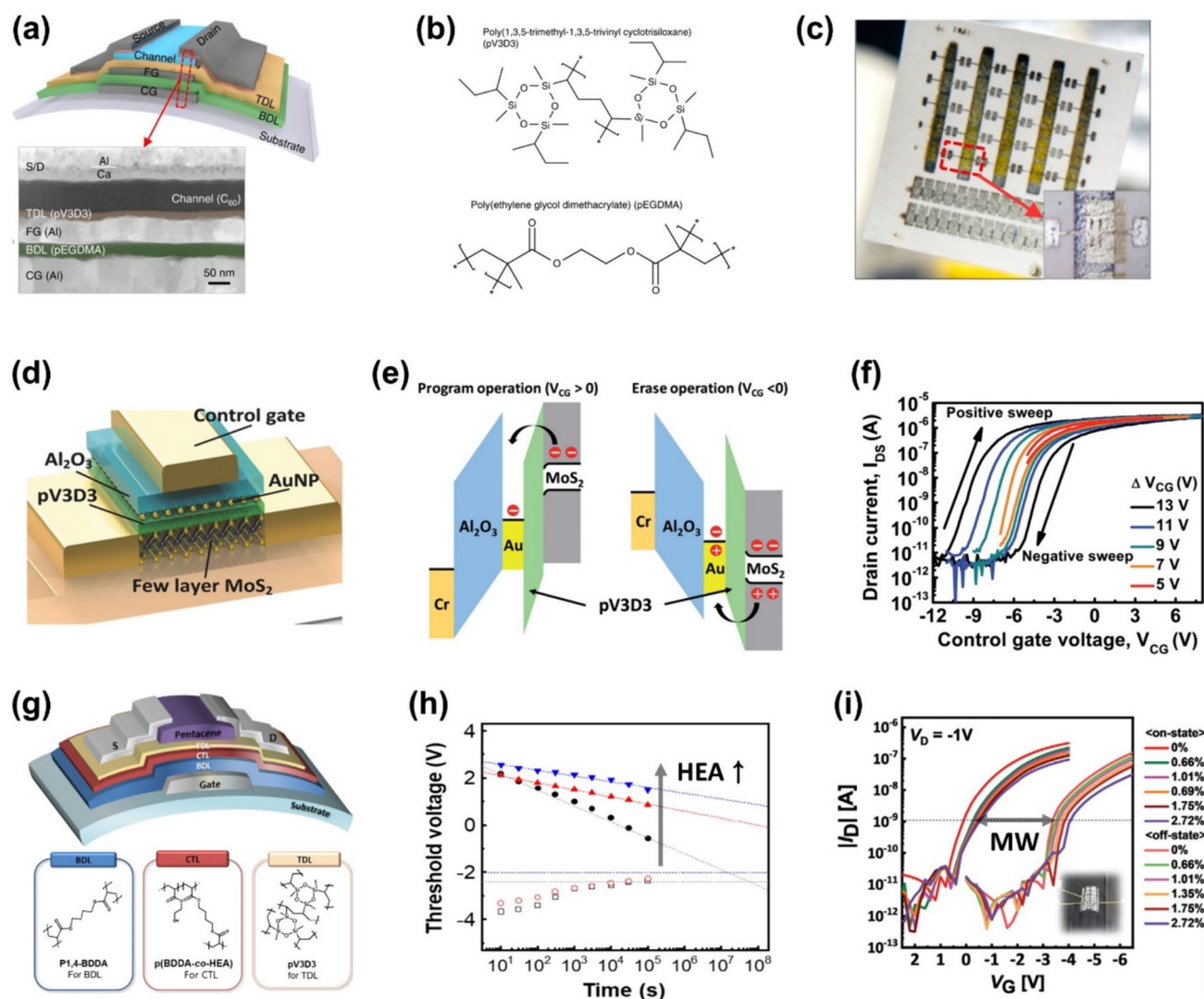


Fig. 6 Charge trapping memory based on vapor-phase synthesized polymer dielectric layer. **a** Schematic illustration of the fabricated memory device, which is composed of a pV3D3 tunneling dielectric, gold nanoparticles (AuNPs) as floating gate, and an Al₂O₃ blocking dielectric layer. **b** Band diagram of programming and erasing operation. **c** Transfer characteristics and memory window with control gate voltage (V_{CG}) sweeping in the negative-to-positive direction and back, with different V_{CG} pulse, with fixed $V_{DS}=1$ V. Reproduced with permission from Ref. [119], Copyright 2017, Springer Nature. **d** Schematic device structure of the proposed flash memory and TEM image. **e** Chemical structures of the pV3D3 and the pEGDMA used

as TDL and BDL, respectively. **f** A photograph of the organic memories deposited on paper substrate. Reproduced with permission from Ref. [125], Copyright 2017, Wiley-VCH GmbH. **g** A schematic diagram of the memory with the trilayer dielectric composed of pV3D3 TDL, p(BDDA-co-HEA) CTL, and pBDDA BDL. Characteristics of the p(BDDA-co-HEA) layer. **h** Memory characteristics of the flexible memory on a PEN substrate under the applied tensile strain of 2.72%. **i** Retention characteristics depending on fraction of HEA in CTL. Reproduced with permission from Ref. [7], Copyright 2020, Wiley-VCH GmbH

them incompatible with the conventional photolithography process, and thus making it difficult to realize robust metal interconnection with multi-layered structure for large-scale ICs [140–145].

The iCVD process, grounded in adsorption chemistry [146, 147], facilitates the use of a shadow mask as an efficient method for the bottom-up deposition of a polymer dielectric layer pattern, enabling patterned interlayer dielectric layer (ILD) without the need for the lithography process

[148]. Park et al. demonstrated a multi-layered organic logic circuit array by employing via-hole-less metal interconnects through the iCVD process [149]. As shown in Fig. 7a, even a fundamental logic component such as an XOR gate necessitates four AND gates, with each AND gate comprising four transistors. Moreover, a seamless interlayer connection is essential for the functionality of the logic gate [52]. Through the application of iCVD process, they effectively constructed an XOR logic gate with an array of 16 OTFTs

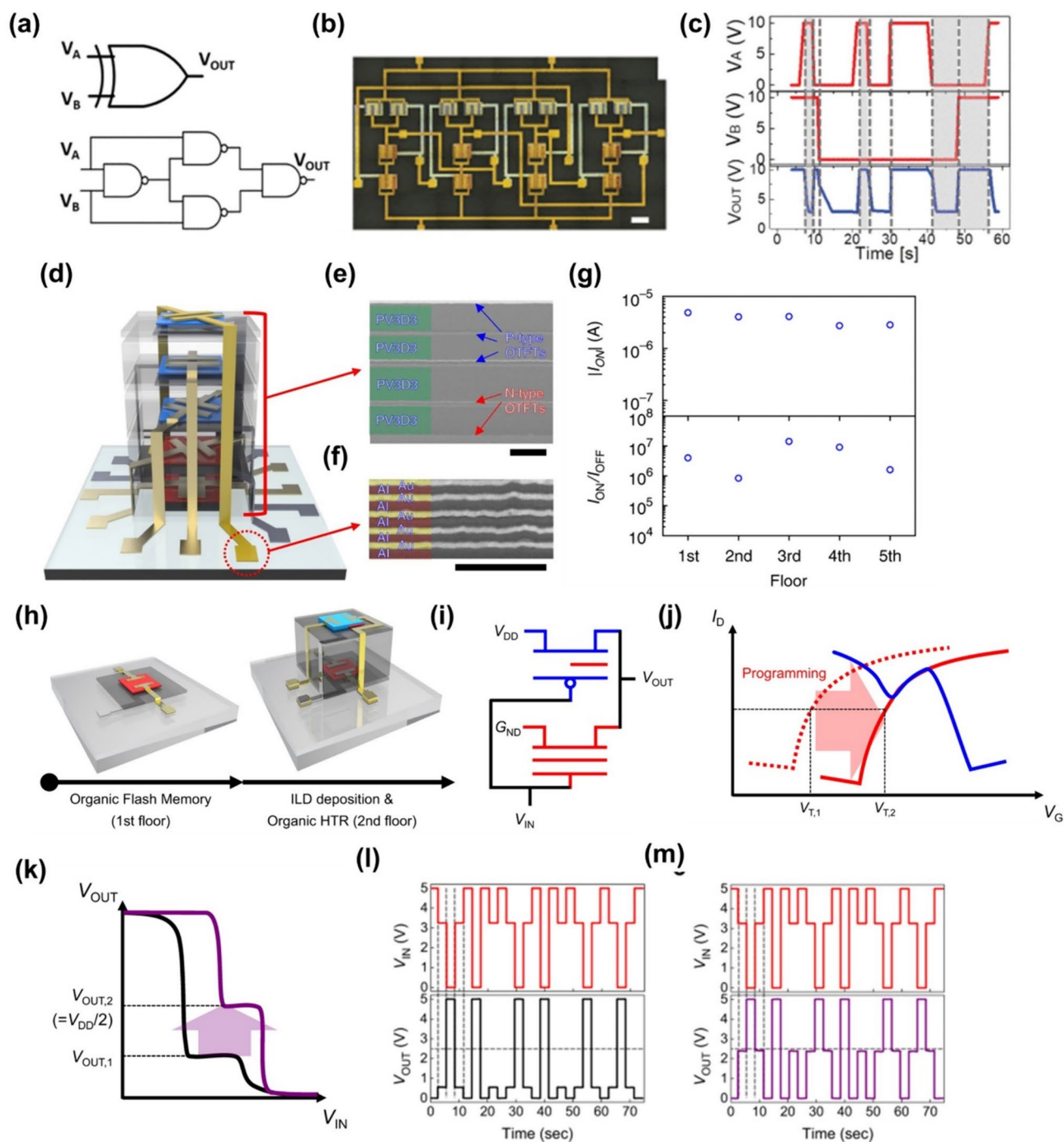


Fig. 7 Demonstration of the OTFT-based logic devices using iCVD polymer dielectric. **a** Circuit symbol and diagram of the XOR circuit. **b** Optical image of the XOR circuit, scale bar, 600 μm . **c** Input and output characteristic of the XOR circuit. Reproduced with permission from Ref. [149], Copyright 2020, IEEE. **d** Schematic of a 5-layer 3D-stacked OTFT. **e** Cross-sectional SEM images of 3D OTFT, scale bar, 1 μm , and **f** interconnected metal electrodes, scale bar, 400 μm . **g** On-current values and on/off current ratio of 5

3D-OTFTs. Reproduced with permission from Ref. [18], Copyright 2019, Springer Nature. **h** 3-dimensional ternary logic inverter (3D T-inverter) fabrication method, and **i** circuit schematic. **j** Operating principle when the organic flash memory is programmed, and **k** corresponding transfer characteristic of the 3D T-inverter. Transient measurement results of 3D T-inverter with **l** pristine and **m** programmed states. Reproduced with permission from Ref. [163], Copyright 2022, Springer Nature

(Fig. 7b). The logic gate operated efficiently, exhibiting full XOR gate features, as corroborated by the transient device signal assessment in Fig. 7c. This research highlighted the possibilities of iCVD technology in intricate circuit layouts.

The iCVD method served as an effective tool for the creation of vertically stacked, high-density integrated circuits through via-hole-less layer interconnects and ILD, which are pivotal for compact device integration without requiring ultra-high line resolution [150–154]. Yoo et al. introduced an innovative method to construct interconnect structures using representative iCVD dielectric-pV3D3, a low- k polymer dielectric with superior insulating characteristics [18]. They achieved successful metal interconnects, where each layer with a 1 μm -thick vertical separation was connected seamlessly with no electrical degradation. Accordingly, they successfully fabricated a vertically stacked OTFT devices in a 5-layer configuration, where the 5 OTFTs were assembled with distinct layer-by-layer separation (Fig. 7d–f). Moreover, owing to the mild process condition of the iCVD process, all 5 OTFTs functioned without any disturbance or variation between layers (Fig. 7g), highlighting the capabilities of iCVD process in vertically stacked, multi-level circuit fabrication.

Along with the vertically stacked logic circuits, multi-valued logic (MVL) circuits garnered huge attention to increase integration density without reducing the size of the unit devices, thereby effectively mitigating the scaling issues of organic semiconductors [155–159]. The MVL circuits utilize more than three logic states and allow for processing dense information more efficiently without requiring intricate circuit architectures [158, 160]. The organic-based MVL has been implemented through the utilization of organic heterojunction transistors (H-TRs), which usually employed heterojunction channel of p-type and n-type semiconductors in a unit transistor [161]. The H-TRs exhibit a unique intermediate negative transconductance (NTC) region in its transfer characteristics, enabling ternary logic operation within the conventional complementary inverter scheme [155, 157, 160, 162]. Using iCVD process, Choi et al. fabricated a vertically stacked ternary logic gate by serially coupling the H-TR with an organic flash memory devices (Fig. 7h, i) [163]. The arrangement allowed for programming of the circuit to attain an ideal intermediate logic state. As demonstrated in Fig. 7j, the transfer curve of H-TRs (blue line) aligned with the non-volatile memory (NVM) transfer curve (red dotted line) to formulate a ternary inverter curve. Adjusting the threshold voltage of the NVM device through programming (red shear line) enables the formation of a comprehensive intermediate logic state, as evidenced in Fig. 7k. They conducted a comprehensive analysis on the transient characteristics of the ternary inverter in both its non-programmed (Fig. 7l) and programmed (Fig. 7m) states, revealing that complex logic circuits such as XOR

or advanced multi-bit logic can be realized using iCVD-enabled MVL memory devices, marking a significant stride in the evolution of future computing technologies.

Beyond the logic devices, research about intelligent devices have also been investigated intensively. Intelligent devices capable of implementing probabilistic reasoning at the hardware level with the iCVD process, as used in simulations for drone flight path planning. Conventional artificial neural networks (ANNs) for deep learning have encountered an overconfidence issue [164, 165], leading to the emergence of Bayesian inference based on probability distributions [166]. In order to perform Bayesian inference, it is necessary to realize hardware capable of executing probability functions, such as Gaussian distribution. However, as shown in Fig. 8a, too many memristors have been required for 'point by point operation' in multiply and accumulation (MAC) operations. To overcome this, intelligent devices that could generate Gaussian distribution through hardware by adjusting ' μ ' (mean) and ' σ ' (standard deviation) were developed based on CMOS circuits [166]. Yet, this approach still required a large number of transistors, and thereby making the approach inefficient in terms of power consumption and large footprint [167, 168]. In response, Lee et al. successfully developed a memory-structured device (anti-ambipolar transistors with p, n junctions serving as channels), which is called Gaussian-like memory transistor (GMT). This device could individually adjust μ and σ at a single device level with separated FGs that allow the independent programming of p, n-type semiconductors [169]. As shown in Fig. 8b, the device was fabricated by inserting gold nanoparticle (AuNP) FGs between a BDL composed of pC1D1 and a TDL composed of pV3D3, within the anti-ambipolar transistor structure. The transfer characteristic of the device exhibited a typical Gaussian-like curve. The same direction of V_{prg} applied to the drain and source could shift the μ of the Gaussian-like transfer curve (Fig. 8c, d), while programming with opposite direction voltage could adjust the σ as designed (Fig. 8e, f). Based on the Gaussian-like transfer curve with independently adjustable μ and σ , a simulation was conducted for path planning after obstacle localization through probabilistic reasoning. (Fig. 8g) This resulted in clear conclusions that closely match the ground pose and conventional models (Fig. 8h).

Conclusion

In summary, the review comprehensively outlines the progress of iCVD polymer dielectrics and their applications in various state-of-the-art electronic devices for last decade. Significant advancements have been achieved since the first demonstration of iCVD polymer gate dielectric layer for high-performance OTFTs. The motivation for advancing

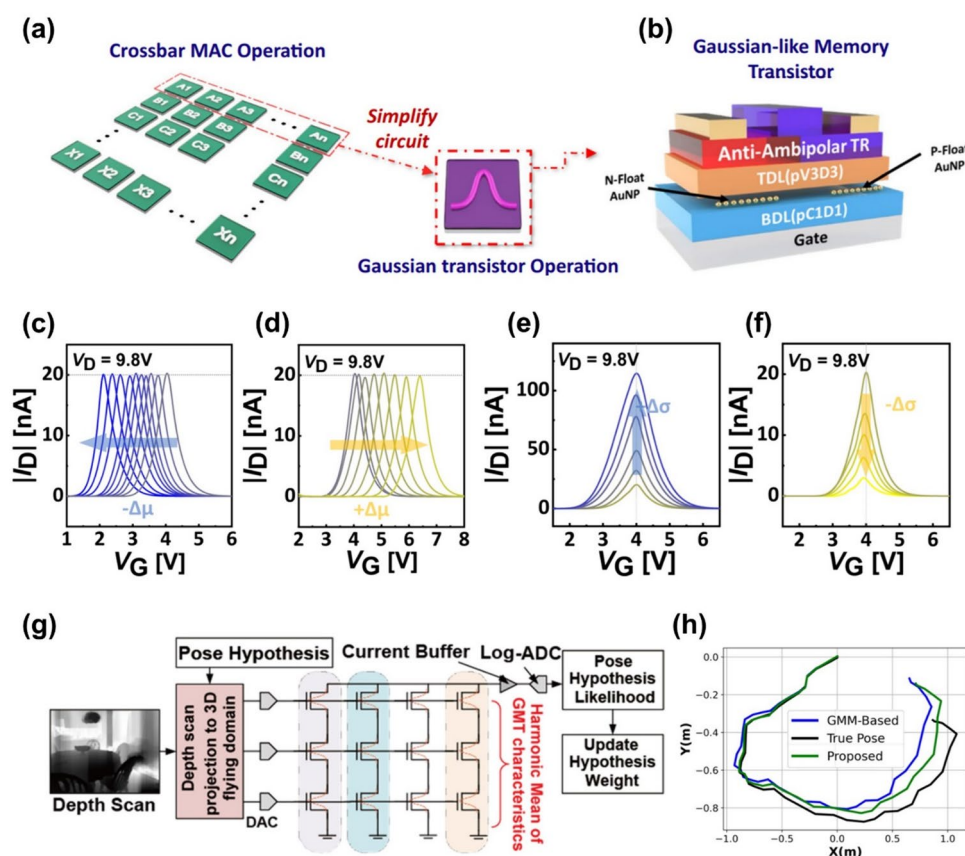


Fig. 8 An intelligent device capable of creating a Gaussian-like distribution-shaped transfer curve using iCVD dielectric layer. **a** A schematic concept of the probabilistic reasoning using a Gaussian-like output transistor compared to the multiply and accumulation (MAC) operation **b** a schematic illustration of a Gaussian-like memory transistor (GMT) device structure. **c** The transfer characteristics (absolute value of drain current ($|I_D|$) versus V_G measured in Case 1 (movement of p-type threshold voltage ($\Delta V_{T,p}$)=movement of n-type threshold

voltage ($\Delta V_{T,n}$) with the p-type programming voltage ($V_{prg,p}$) < 0 and **d** $V_{prg,p}$ > 0. **e** The transfer characteristics measured in Case 2 ($\Delta V_{T,p} = -\Delta V_{T,n}$) with the condition of $V_{prg,p}$ > 0 and **f** $V_{prg,p}$ < 0. **g** Mapping of robotic pose localization to Gaussian-like memory transistor (GMT) array **h** Drone position tracking and comparison to ground truth and conventional model. Reproduced with permission from Ref. [169], Copyright 2024, Springer Nature

iCVD polymer dielectrics lies in their low surface roughness, conformal coverage, damage-free nature, and robust insulating characteristics, with film thicknesses ranging from several nanometers to micrometers, which are not easily attainable through conventional solution-based polymer dielectrics. In particular, the iCVD process enables precise control over the thickness as well as chemical composition of the synthesized polymer while preserving the intrinsic functionality of each monomer. Accordingly, beyond robust insulating performance of the ultrathin polymer dielectrics, the iCVD process allows for various functional polymer dielectrics such as high- k polymer dielectrics, electron-withdrawing or electron-donating dielectrics, and stretchable dielectrics by imparting unique properties such as high polarity, and mechanical stability against external forces. Such multi-functional polymer dielectrics have driven the realization of various organic electronic devices and logic circuits, including flash memory, RRAM, multi-stage

organic logic circuits, vertically stacked logic circuits, and multi-valued logic circuits.

Given the increasing interest in utilizing lightweight, flexible, and stretchable materials for user-friendly devices compatible with various form factors, there will be continuous and extensive exploration into innovative fabrication techniques for polymer dielectrics that can enhance the performance metrics of organic electronics while fully leveraging the deformable substrates such as paper and plastic. Looking ahead, we believe that continuous research and innovation in iCVD polymer dielectric layers promise to further expand the application of organic electronics, driving the development of future low-power, flexible/stretchable, and wearable electronics.

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Data Availability Data sharing is not applicable to this article as no new data were created or analyzed in this study.

Declarations

Conflict of Interest The authors declare no competing financial interest.

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